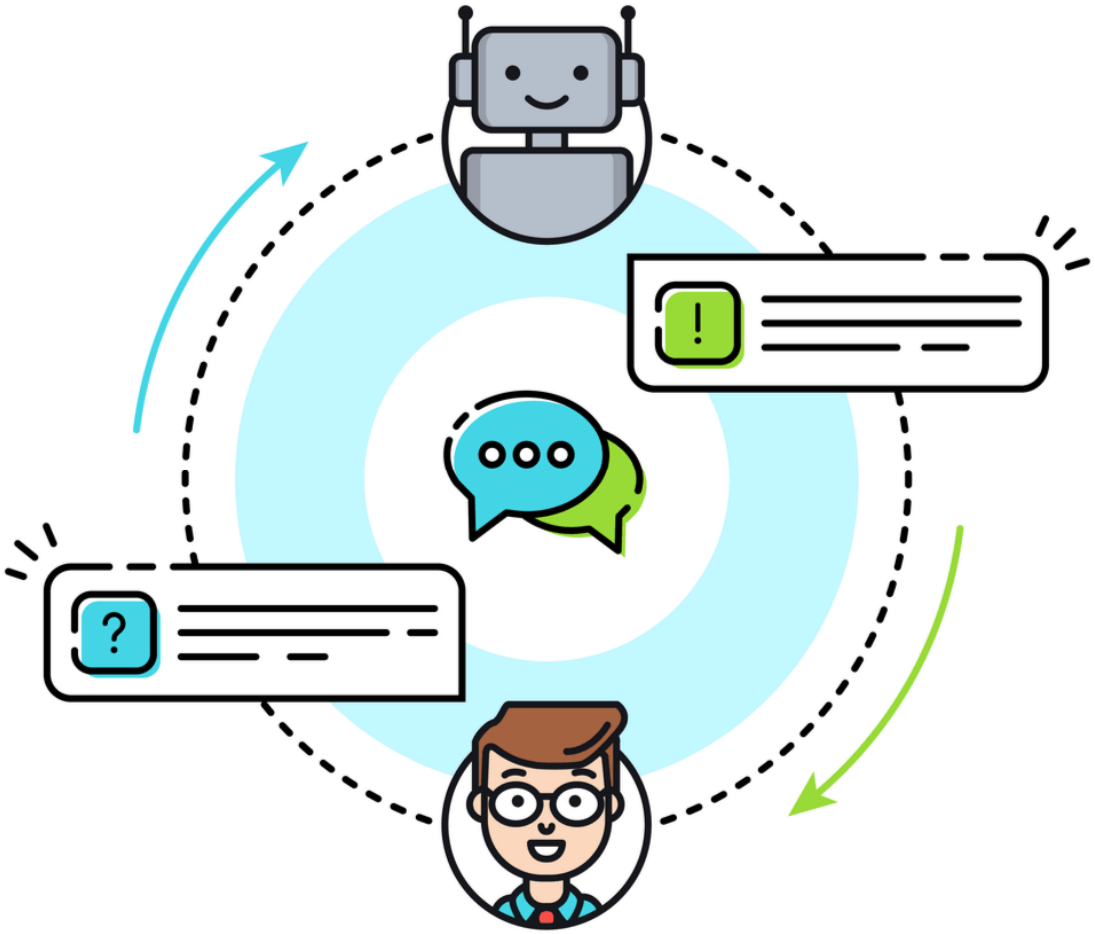
INTRO TO NLP

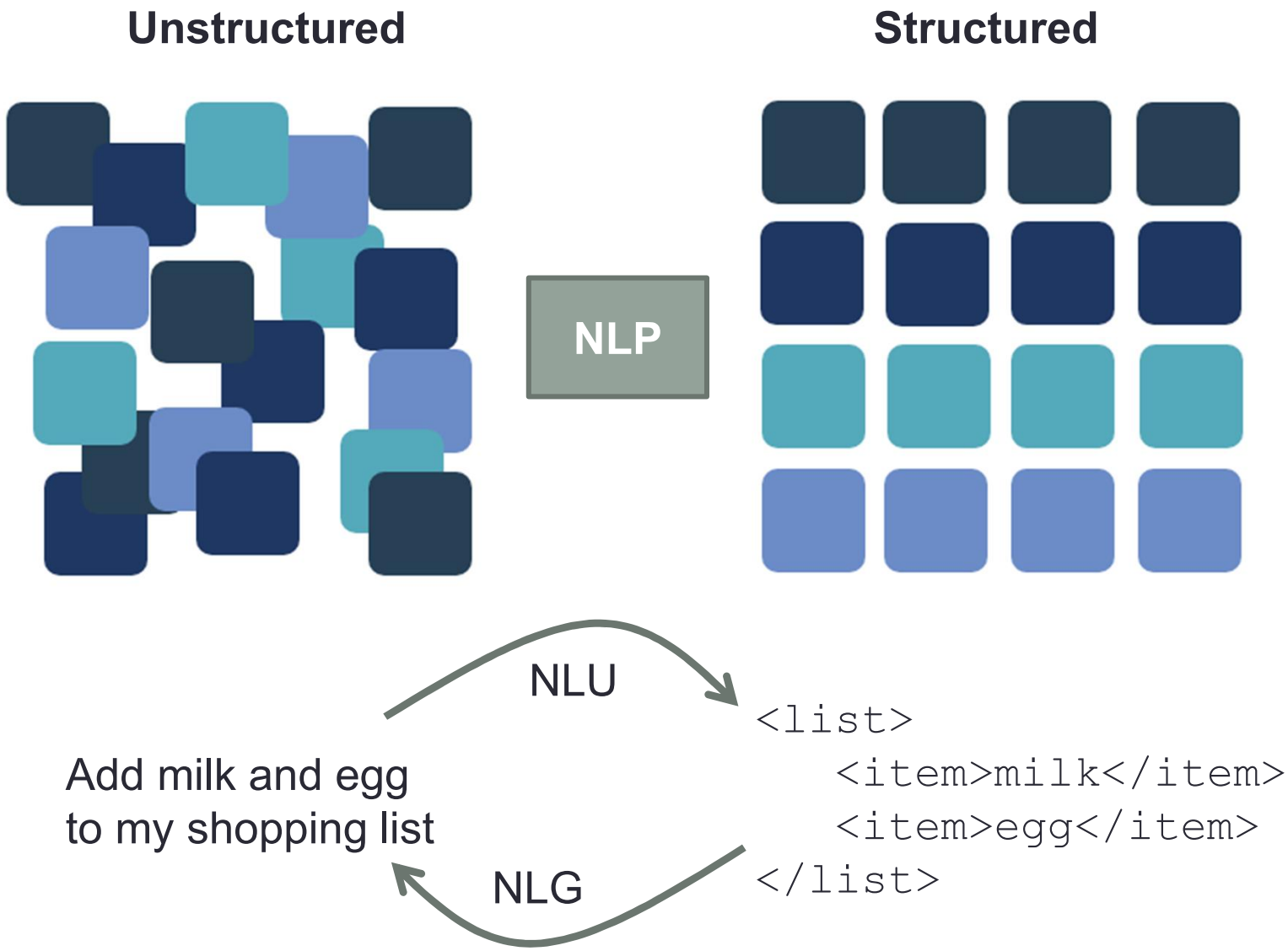
Prepared by Dr. Valy Dona

What is NLP?

- NLP: Natural Language Processing



What is NLP?


Unstructured
Structured
NLP
NLU
NLG
Add milk and egg
to my shopping list
<list>
<item>milk</item>
<item>egg</item>
</list>

Python Environment and Libraries

- Python Review (from Standford CS231n by Justin Johnson):
  - https://cs231n.github.io/python-numpy-tutorial/
- Python Environment: Google Colab (https://colab.research.google.com)
- Libraries:
  - Regular Expression Operations (re)
    - https://docs.python.org/3/library/re.html
  - Natural Language Toolkits (NLTK)
    - https://www.nltk.org/

REGULAR EXPRESSIONS

Prepared by Dr. Valy Dona

Regular Expressions

"A formal language for specifying text strings"

- Character classes

Pattern	Matches
`.`	any character except newline
`\w \d \s`	word, digit, whitespace
`\W \D \S`	not word, digit, whitespace
`[abc]`	either a, b, or c
`[^abc]`	not a, b, or c
`[a-g]`	character between a & g

Regular Expressions (cont.)

- Quantifiers and Alternation

Pattern	Matches
`a* a+ a?`	0 or more, 1 or more, 0 or 1
`a{5} a{2,}`	exactly five, two or more
`a{1,3}`	between one & three
`a+? a{2,}?`	match as few as possible
ab|cd	ab or cd

Regular Expressions (cont.)

- Escaped Characters

Pattern	Matches
`\. \* \\`	escaped special characters
`\t \n \r`	tab, linefeed, carriage return

- Anchors

Pattern	Matches
`^abc$`	start / end of the string
`\b`	word boundary

Regular Expressions (cont.)

- Groups and Lookaround

Pattern	Matches
`(abc)`	capture group (useful with **replace**)
`\1`	backreference to group #1
`(?:abc)`	non-capturing group
`(?=abc)`	positive lookahead
`(?!abc)`	negative lookahead

Exercise

Write a regular expression to match each of the following patterns:

1. Any character that is not a word character nor a white space
2. String of letters whose length is at most 3
3. String of digits whose length is at least 3
4. String of word characters containing at least one a and one b
5. Anything enclosed by square brackets
6. String of word characters whose the 5th character from the right end is a digit
7. Date format: yyyy-mm-dd

Exercise

Write a regular expression to match each of the following patterns:

8. String of 10 digits that starts and ends with the same 3-digit sequence
9. Password satisfying the following conditions:
   - at least 1 lowercase letter
   - at least 1 uppercase letter
   - at least 1 digit
   - six characters or more

References

- Daniel, J., & James, H. M. (2021). Speech and Language Processing.
- Khanna, C. (August, 2021). Byte-Pair Encoding: Subword-based tokenization algorithm. Retrieved from https://towardsdatascience.com/byte-pair-encoding-subword-based-tokenization-algorithm-77828a70bee0